Modeling Spatial Dependence: A Simulation-Based Comparison of Parametric and Semi-Parametric Approaches

Alejandro M. Ouslan
Supervisors: Dr. Raúl Macchiavelli, Dra. Santana,

Dr. Hernández, Dr. Rivera Santiago

Seminario Interuniversitario de Investigación en Ciencias Matemáticas

February 27, 2026

Presentation Overview

1. Introduction
2. Production Function
3. Production Function Cont.
4. Parametric Spatial Models
5. The Spatial Weights Matrix

Why Spatial Dependence Matters

- Regional data often exhibit spatial dependence.
- Example: Puerto Rico municipalities.
- Productivity in one region may depend on neighboring regions.
- Violates classical i.i.d. assumption.

Production Function

$$Y = AK^{\alpha}L^{\beta}$$

$$\log(Y) = \log(A) + \alpha \log(K) + \beta \log(L) + \epsilon; \epsilon \sim N(0, \sigma)$$

- Y is the total output (real GDP) produced.
- A is total factor productivity (TFP), capturing the efficiency with which inputs are used.
- K is the quantity of physical capital used in production.
- L is the quantity of labor employed.

Production Function

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$$\log(Y) = \log(A) + \alpha \log(K) + \beta \log(L) + \epsilon; \epsilon \sim N(0, \sigma)$$

- α is the output elasticity of capital
- β is the output elasticity of labor
- i municipalities id
- t time period

Spatial Autoregressive (SAR) Model

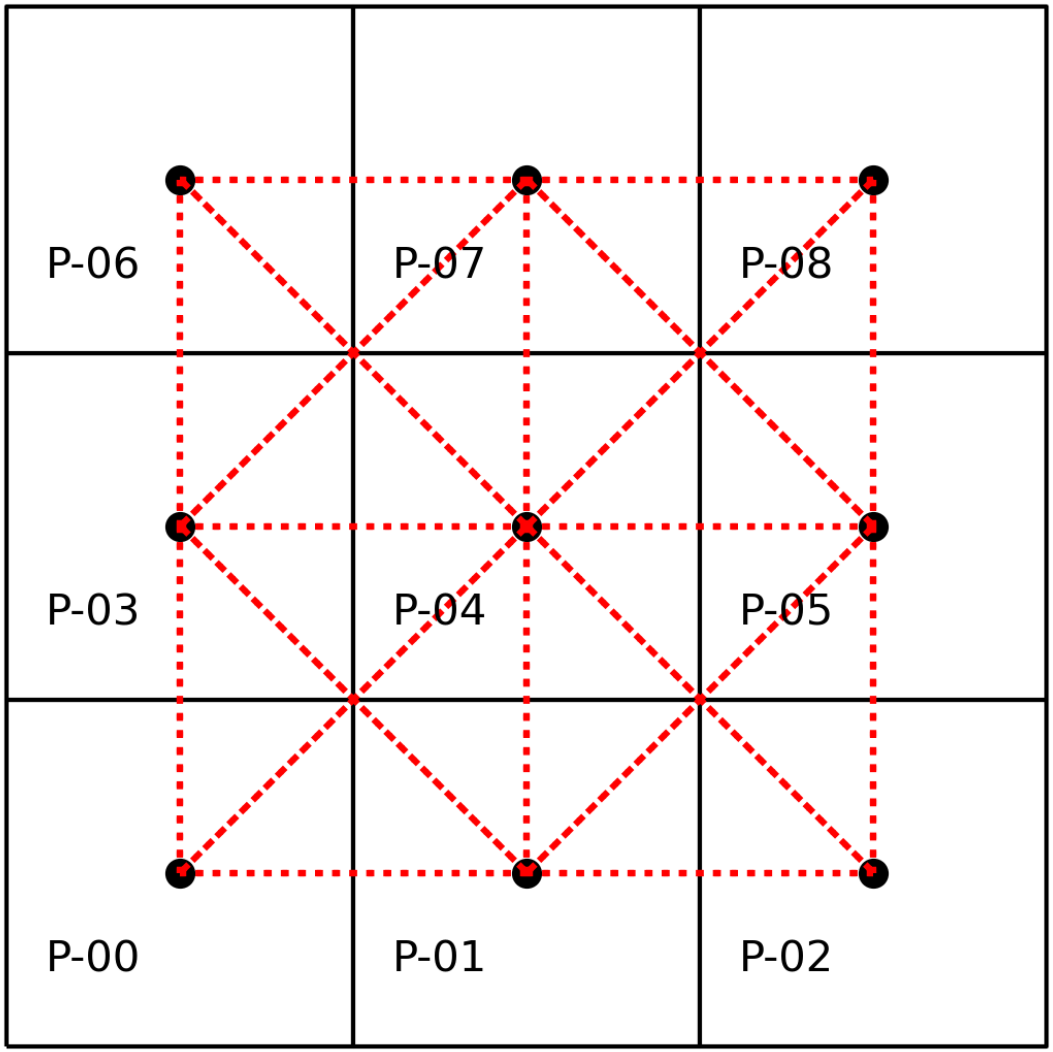
$$y = \rho W y + X\beta + \varepsilon$$

- ρ measures spatial dependence.
- W defines neighborhood structure.
- Sensitive to correct specification of W.

Defining W: The Core Challenge

- W is assumed known in theory.
- In practice, it must be chosen.
- No universally accepted selection rule.
- Misspecification leads to biased estimates.

Example of the Weights Matrix (Queen)



Example of the Weights Matrix (KNN)

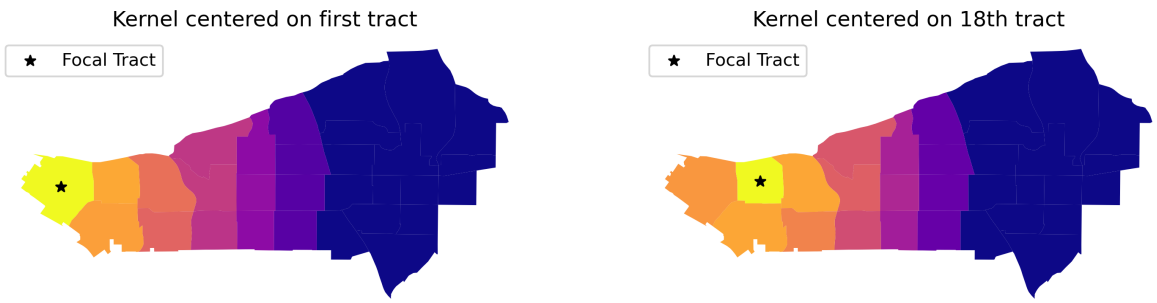


Figure: K-Nearest Neighbors (KNN) Model

Tensor Product Spatial Smoother

$$\log(Y_{it}) = \alpha + \beta_1 \log(K_{it}) + \beta_2 \log(L_{it}) + f(\text{lat}_i, \text{lon}_i) + \varepsilon_{it}$$

Key Idea

No need to specify W. Spatial effects captured flexibly using splines.

Simulation Results (1000 Runs)

Variable	Rook	Queen	Tensor	OLS
Intercept	3973.3	0.40	8252.6	7177.3
X_1	0.74	0.0032	0.21	1.22
X_2	0.90	0.0032	0.26	1.57
X_3	1.04	0.0031	0.27	1.92

Finding: When W is misspecified, the tensor approach is substantially more robust.

Main Conclusions

- Parametric models are highly sensitive to W.
- Misspecification leads to large estimation errors.
- Semi-parametric tensor approach provides robust spatial control.
- Useful in contexts like Puerto Rico where W is uncertain.